utility10_01-seeti-counter

Seeti Counter: manufacturing manual

> **Status: design only.** Nothing here has been built or tested as hardware. This manual tells you exactly what to make, buy, assemble, program and test, and how to accept or reject a unit; expect to correct it after the first prototype run (5 units).

Product summary. A heat-shielded clip puts a small microphone about 10 cm from the whistle. A Goertzel filter looks for the whistle's narrow tone; a whistle must last 0.4 s to count. After N whistles (button picks 1–6) a buzzer rings 5 times.

Product line / price in the plan: Tier A · low-voltage electronics · prototype Rs 1080 · price Rs 999. Status: designed and verified as far as a computer can; no prototype built, nothing field-tested.

1. What you are building: deliverables

Deliverable	Where	State
PCB: Gerbers, BOM, CPL, KiCad project	`utility10/pcb/release/seeti/`	**order-ready: every automatic check passes** (DRC errors 0, netlist match True)
Firmware (`seeti_prod.hex`)	`utility10/firmware/release/4MHz/seeti_prod.hex`, source `utility10/firmware/src/main_board.cpp` (seeti: `seeti_prod.cpp`), drivers `board_drivers.h`, engine `utility10/firmware/lib/utilcore.c`	engine host-tested (115 checks); board firmware compiles but is **not yet run on hardware** (see section 6)
Enclosure STL (prototype shell)	`utility10/01-seeti-counter/enclosure_body.stl`, `enclosure_lid.stl` and the board's `4_enclosure/`	prototype shell only
Mechanical / bought-in parts	section 3	listed
Test sheet	`manuals/templates/eol_test_sheet.md`	template

2. Before you start

- Read `manuals/STANDARD_PROCEDURES.md` (sections A-F, J, K).
- Tools: soldering iron, flux, multimeter, USBasp programmer + avrdude, calipers, 3D printer or print service, bench supply with current limit.
- Power source for this product: **2 x AAA**.

3. Bill of materials

3a. Circuit board parts (32 parts, 4 test points)

SMD parts marked `SMT (JLC)` are assembled by the fab from the JLC BOM; parts marked `hand/THT` are soldered by you. Every LCSC number was checked against an LCSC/JLCPCB listing on 2026-11-01; re-check stock on the day you order.

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
U2	MCP6001	SOT-23-5	MCP6001T-I/OT	C116490	SMT (JLC)
J1	BATT 2xAAA	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
SW1	Power	SW_SPDT_PCM12	C&K PCM12SMTR	C221841	SMT (JLC)
Q2	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT
J3	MIC electret	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
J4	UART	PinHeader_1x04_P2.54mm_Vertical	ZHOURI 2.54-1*4	C5116483	hand/THT
BZ1	Buzzer 3V	Buzzer_12x9.5RM7.6	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	hand/THT
Q1	MMBT3904	SOT-23	MMBT3904	C20526	SMT (JLC)
D2	1N4148WS	D_SOD-323	1N4148WS	C2128	SMT (JLC)
SW2	Target	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
D1	Green	LED_0603_1608Metric	Everlight 19-217/GHC-YR1S2/3T (LCSC showed it out of stock on the search day: check)	C72043	SMT (JLC)
R1	2.2k	R_0603_1608Metric	0603 2.2k 1%	C4190	SMT (JLC)
R2	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R3	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R4	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R5	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
R6	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
R8	33k	R_0603_1608Metric	UNI-ROYAL 0603WAF3302T5E	C4216	SMT (JLC)
R9	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
C1	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C4	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
C5	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
C6	220n	C_0603_1608Metric	Samsung CL10B224KB8NNNC X7R 50V	C64705	SMT (JLC)
C7	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
C8	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
R10	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C9	22n	C_0603_1608Metric	Samsung CL10B223KB8NNNC X7R	C21122	SMT (JLC)

3b. Bought-in and mechanical items (estimates for about 2,000 units; replace with quotes)

Item	Rs
MCU + crystal	45
Electret mic amp module	40
Buzzer	12
PCB	30
2×AAA holder + switch	18
Heat-resistant clip/magnet mount	40
Enclosure	50
Packaging + Hindi/Telugu manual	30
Assembly + test	40

4. Step 1: fabricate the PCB

Upload `utility10/pcb/release/seeti/1_fabrication/seeti_counter_gerbers.zip` with the settings in STANDARD_PROCEDURES section A. Finish: HASL lead-free. Board size and layer list are in `utility10/pcb/release/seeti/README.md`.

5. Step 2: assemble the board

1. Order SMT assembly as in STANDARD_PROCEDURES section B (or hand-solder the SMD parts in section D order).
2. Solder the through-hole parts listed `hand/THT` in 3a.
3. Clean, inspect under magnification, and compare with `2_assembly/*_assembly_top.pdf`.

6. Step 3: program the microcontroller

1. Connect a USBasp to the 6-pin ISP header J2 (pin 1 = square pad) with the board powered (2 x AAA).
2. Set fuses: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFE:m` (internal 8 MHz RC, CKDIV8 off, brown-out 1.8 V; the firmware then divides the clock to 4 MHz (2 x AAA supply may fall to 2.0 V)).
3. Flash and verify: `avrdude -p m328p -c usbasp -U flash:w:utility10/firmware/release/4MHz/seeti_prod.hex:i` then `-U flash:v:utility10/firmware/release/4MHz/seeti_prod.hex:i`.
4. Remove the programmer, power-cycle the board, and confirm the start-up behaviour (section 9).

If you change the supply type (for example USB instead of batteries) use the hex built for that clock (`4MHz` for 2 x AAA, `8MHz` for every other tool).

Firmware pin map (generated from the PCB spec, so firmware and board agree):

Arduino pin	ATmega pin	Net
D7	11	LED
D9	13	BUZ
A0	23	ADC_IN
D0	30	RXD
D1	31	TXD
D2	32	BTN

What is verified: the control logic (`utilcore.c`) passes 115 host tests. **What is not:** the display, OLED, RTC and HX711 drivers and the button flows in `main_board.cpp` were written from datasheets and only compile-checked. Budget bring-up time for them on the first prototype.

7. Step 4: electrical bring-up

1. Insert 2 x AAA with switch OFF; measure BAT_P to GND (J1 pins) = 2.8-3.2 V.
2. Switch ON: TP1 (+3V0) = battery minus about 0.05 V; no heating.
3. VMID (R2/R3 junction) = VCC/2 +-0.05 V; TP3 (MIC_OUT) = VCC/2 +-0.1 V with no sound (same DC at the ADC pin through R10).
4. Flash firmware over ISP (J2): `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFE:m` (internal 8 MHz RC, CKDIV8 off, brown-out 1.8 V; the firmware then divides the clock to 4 MHz).
5. Play a 2.8 kHz tone from a phone at 10 cm: TP3 swings about +-100 mV peak and the firmware counts whistles.

8. Step 5: sensors, cables and mechanical assembly

- **Heat-resistant microphone clip.** Solder the electret capsule (J3, 2-pin JST-PH lead, + toward the marked pin) to a 30 cm 2-core lead. Fix the capsule in the clip behind a stainless-steel mesh window so steam cannot reach it. The clip must hold the capsule 10 cm from the cooker vent and be rated to 85 C (use a high-temperature ABS or metal spring clip; do not use a PLA part near the vent).
- Add a magnet or hook to the unit's back so it stays on the hood; the electronics unit itself must stay at least 10 cm from the vent.

9. Step 6: enclosure and final assembly

1. Print the shell (STANDARD_PROCEDURES section G). Enclosure: `enclosure_body.stl` / `enclosure_lid.stl` (70×45×22 mm, 2.4 mm walls, watertight): a simple prototype shell; production tooling needs a proper design.
2. Place the board on the standoffs (4 mm), fasten with M2.5 screws (6 mm), route cables through the cut-outs, close the lid.
3. Label: serial number, rev, battery type/polarity, warnings.

10. Step 7: calibration

- Whistle frequency is an assumption (2.8 kHz in the tests). Record 5 real cookers of the brands you target, find the dominant whistle frequency and amplitude with a phone spectrum app, and set the filter frequency in `firmware/lib/utilcore.c` (`seeti_*`) before the production run. Re-run the 115 host tests (`make test`).

11. Step 8: end-of-line functional test (every unit)

1. Play a 2.8 kHz tone from a phone speaker at 10 cm for 1 s: the counter increments; play white noise: it must not count.
2. Press the button to select 1, 3 and 6 whistles; after the selected count the buzzer rings 5 times.
3. Battery: measure the supply current at 3.0 V; it should be a few mA (estimate 3.6 mA); a unit above 10 mA fails.

Record results on `manuals/templates/eol_test_sheet.md`. Accept only if every line passes.

12. Packaging, documents, compliance

- In the box: unit, batteries or note on the required adapter, mounting accessories, quick-start card (`manuals/templates/quick_start_card.md`), warranty card.
- Compliance items to verify before sale: Battery safety, BIS requirement for any bought-in adapter, EMC, e-waste EPR, Legal Metrology labelling; no safety claims..
- Labelling: STANDARD_PROCEDURES K.

13. Troubleshooting (product specific)

Symptom	Cause / action
Counts nothing	microphone polarity reversed or gain too low: check TP3 swing at a 2.8 kHz tone, about +-100 mV
Counts noise	threshold or filter frequency wrong for this microphone placement; see the limit in the README (amplitude 300 against noise 1200 counted 0 of 3)

General issues: STANDARD_PROCEDURES section L.

14. Known limits and risks

Synthetic kitchen audio (noise + 100 Hz hum): 3/3 whistles counted, 0 counts for 100 ms clinks, 0 for loud noise alone, 0 for a different-frequency tone, 3/3 with whistle 1.5k vs noise 1.5k amplitude. **Limit found:** a whistle at amplitude 300 against noise 1200 counted 0 of 3, so mic placement and gain matter. The whistle frequency (2.8 kHz here) is an assumption: **record 5 real cookers and set it**.

Whistle frequency varies by cooker brand; steam moisture on the mic; kitchen heat; false counts from a second cooker.

Nothing in this manual replaces testing with real users, a design review by a PCB engineer, and the compliance checks above.